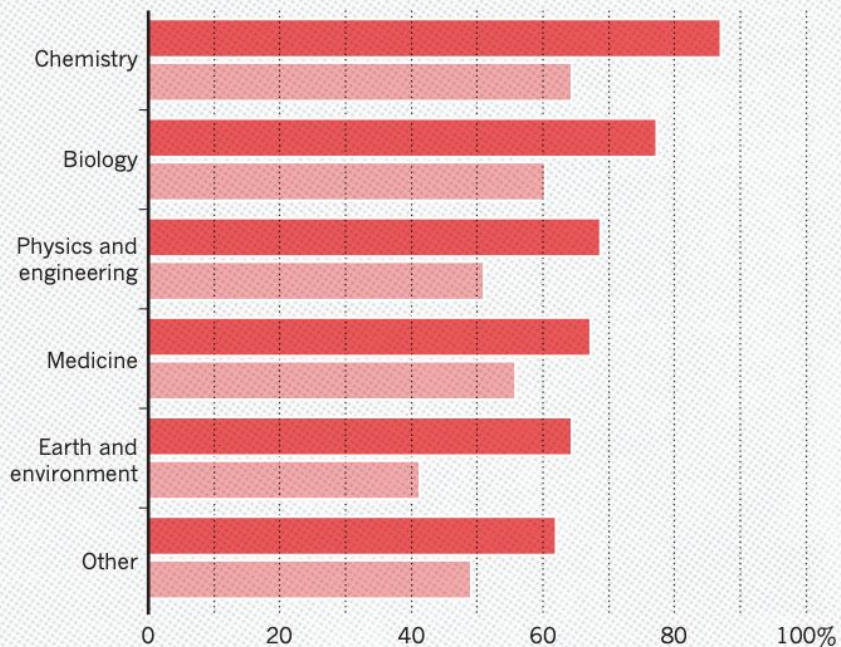


Software Engineering for Scientist

HAVE YOU FAILED TO REPRODUCE AN EXPERIMENT?

Most scientists have experienced failure to reproduce results.

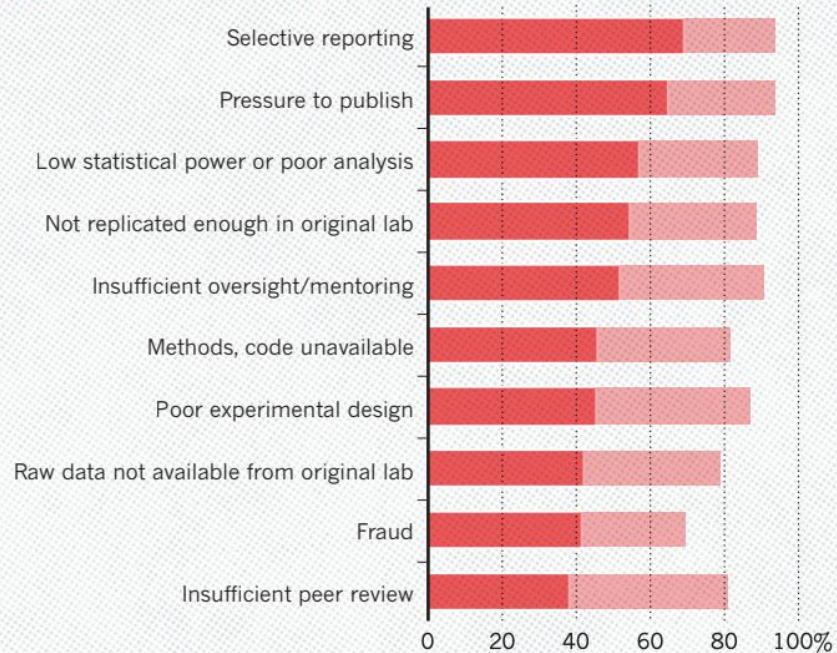
● Someone else's ● My own



WHAT FACTORS CONTRIBUTE TO IRREPRODUCIBLE RESEARCH?

Many top-rated factors relate to intense competition and time pressure.

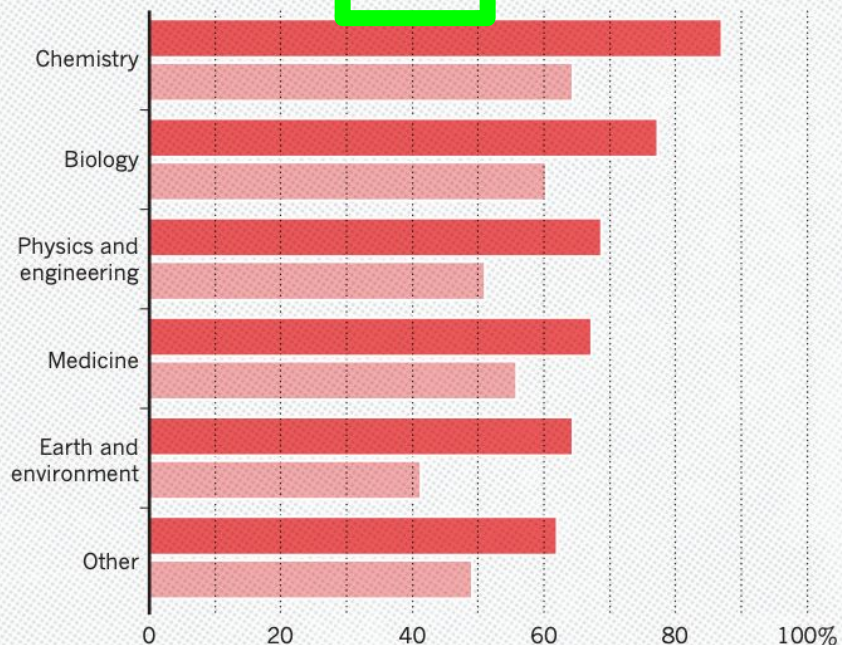
● Always/often contribute ● Sometimes contribute



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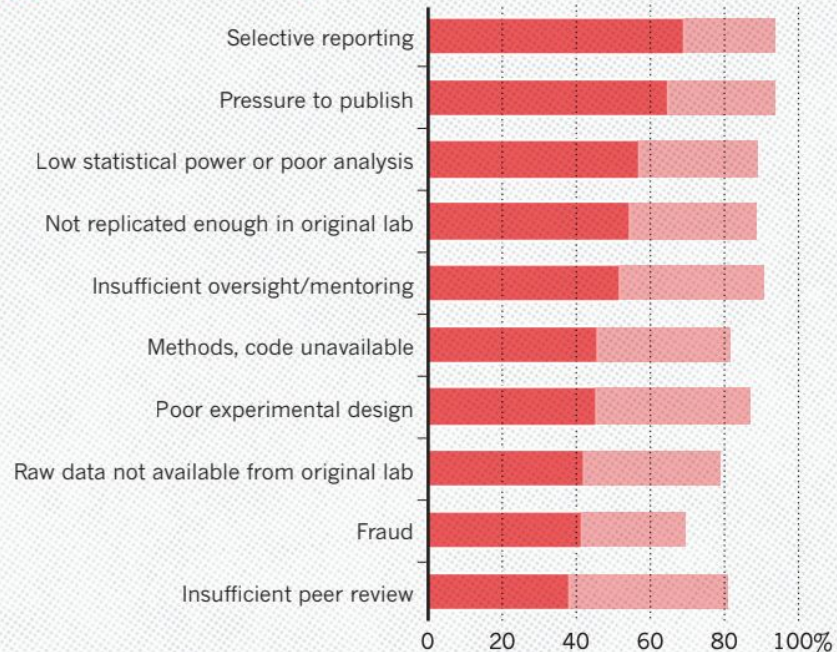
● Someone else's ● My own



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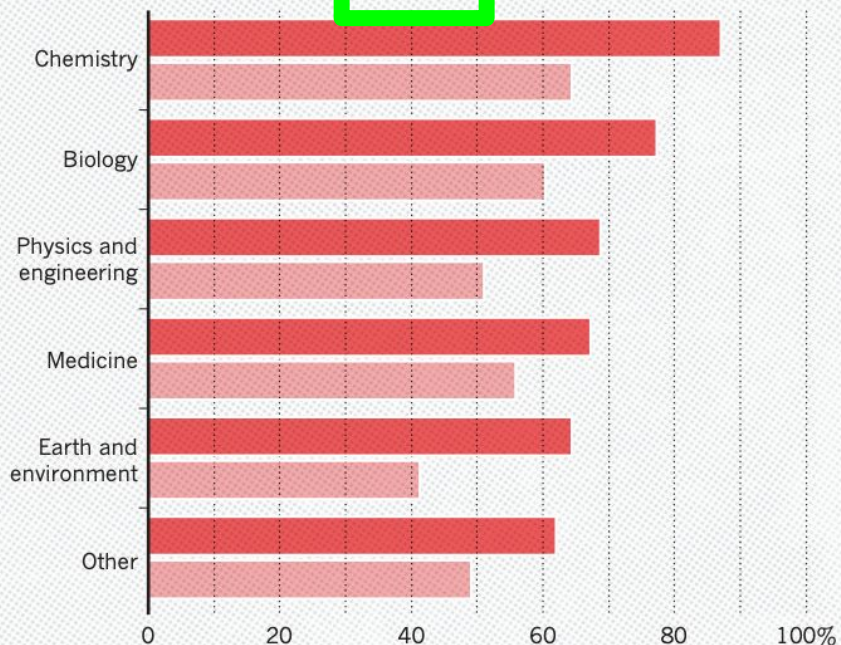
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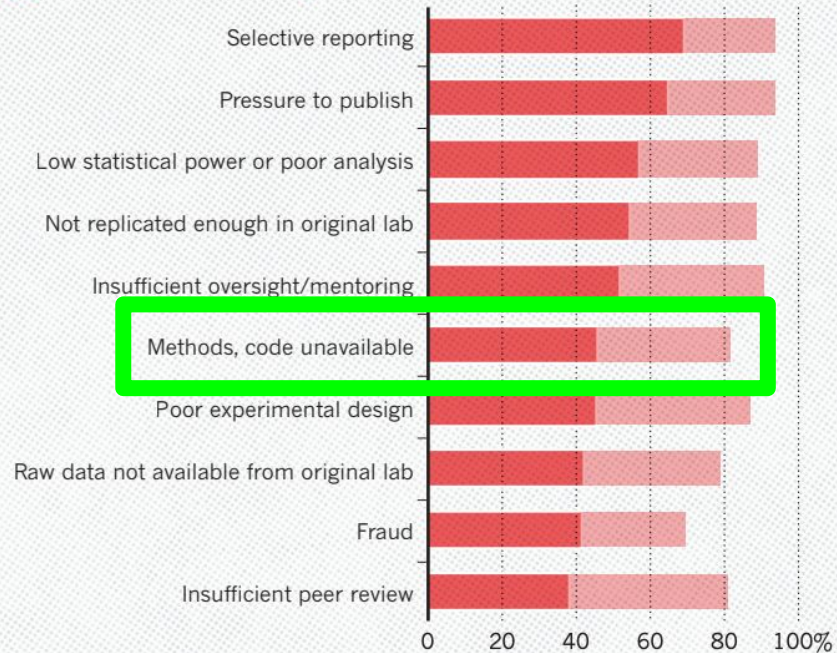
● Someone else's ● My own

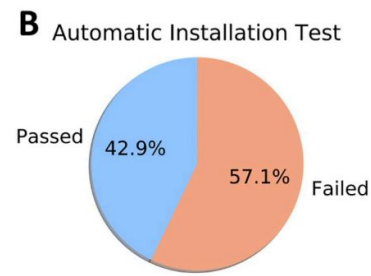
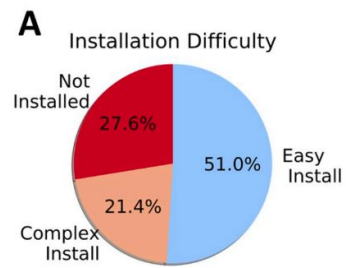


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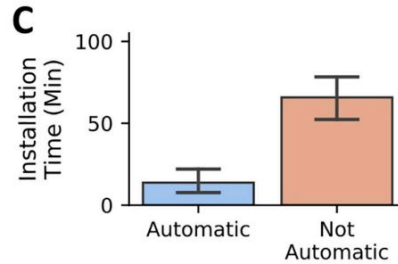
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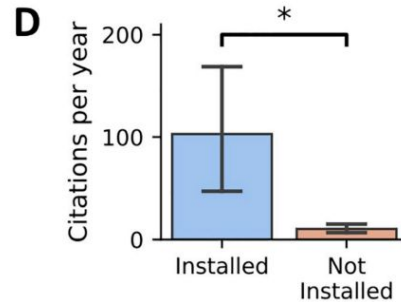




Installation is hard



and slow



and it affects impact

A random forest-based framework for genotyping and accuracy assessment of copy number variations

Xuehan Zhuang, Rui Ye, Man-Ting So, Wai-Yee Lam, Anwarul Karim, Michelle Yu, Ngoc Diem Ngo, Stacey S Cherny, Paul Kwong-Hang Tam, Maria-Mercè Garcia-Barcelo, Clara Sze-man Tang, Pak Chung Sham

<https://github.com/sunnyzxh/CNV-JACG>

```
&showLog("Calculating het probability for $sample...");  
`/home/yerui/miniconda2/bin/perl $home/bin/Het-prob.pl -f $ref -m $snps -r $precnv -i $bam -o $outdir/$precnvbase.het.prob`;
```

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Computer Science 1: Starting Computing



Computer Science 2: Data Structures

Computer Systems

Discrete Structures

Software Development Methods and Tools

Principles of Programming Languages

Algorithms

Linear Algebra

Introduction to Data Science

CSCI Senior Capstone



Foundations of Software Engineering

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Foundations of Software Engineering



LLMS
TO THE
RESCUE!



COLORADO
BOULDER



Writing Reproducible Code

300 prompts like:

Create a Python web scraper that extracts product information (name, price, rating) from an e-commerce website.

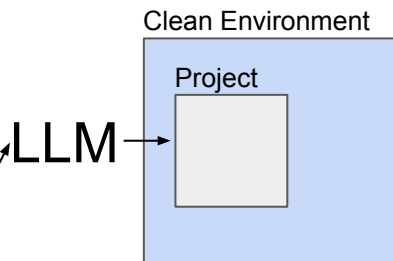
Requirements:

- Use requests and BeautifulSoup4 libraries
- Handle pagination (scrape at least 3 pages)
- Save results to CSV file
- Include error handling for network failures
- Add user-agent headers to avoid blocking

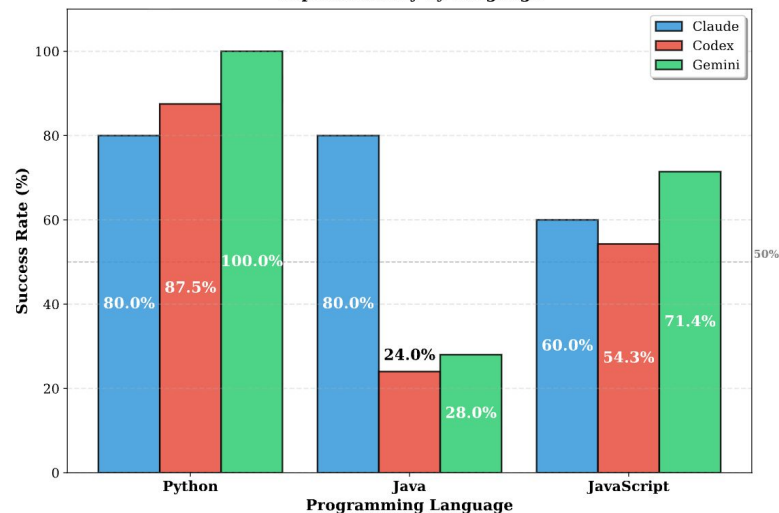
IMPORTANT: Provide EVERYTHING needed for reproduction:

- Complete working Python code
- Complete requirements.txt with ALL dependencies and versions
- Brief usage instructions

Make it 100% reproducible in a clean environment.



Reproducibility by Language



Writing Reproducible Code

300 prompts like:

Create a Python web scraper that extracts product information (name, price, rating) from an e-commerce website.

Requirements:

- Use requests and BeautifulSoup4 libraries
- Handle pagination (scrape at least 3 pages)
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IMPORTANT: **Provide EVERYTHING** needed for reproduction:

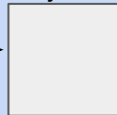
- Complete working Python code
- Complete requirements.txt with **ALL dependencies** and versions
- Brief usage instructions

Make it **100% reproducible** in a clean environment.

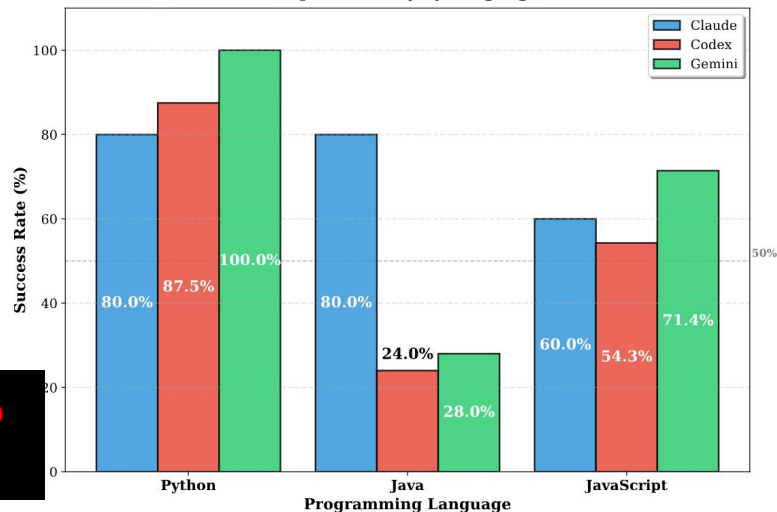
LLM

Clean Environment

Project



Reproducibility by Language

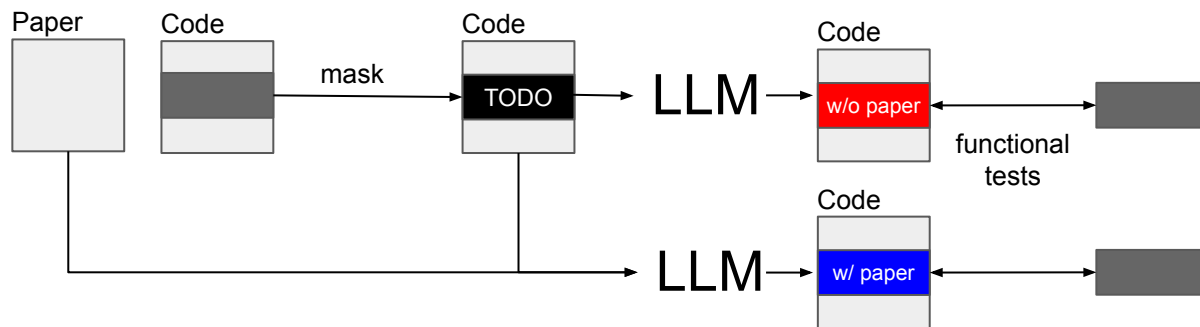


Cannot rely on the LLM to solve your dependencies

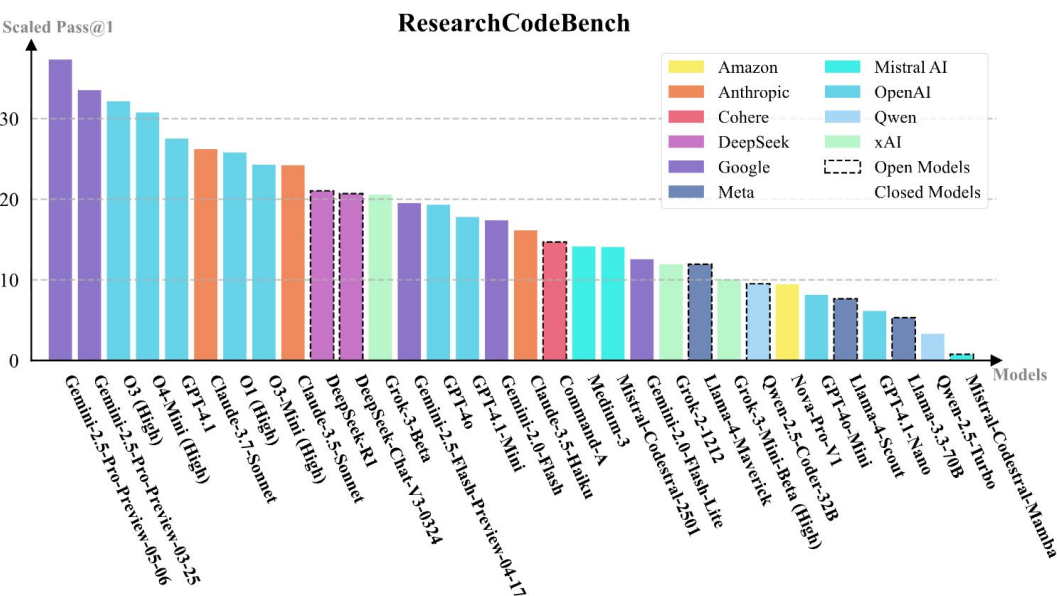


Implementing Ideas

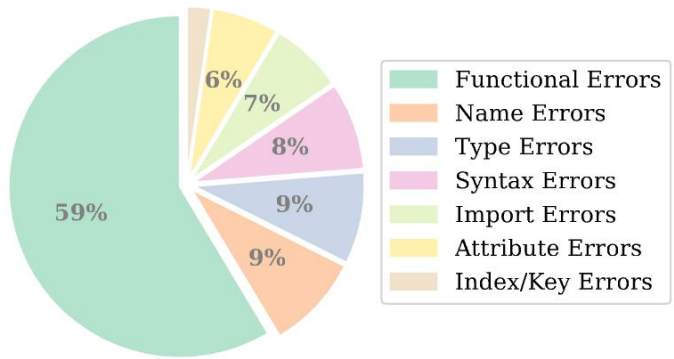
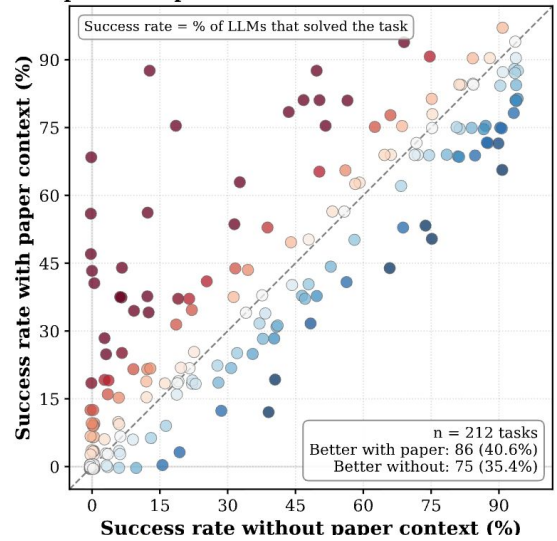
Advantage Alignment Algorithms
Duque, 2025



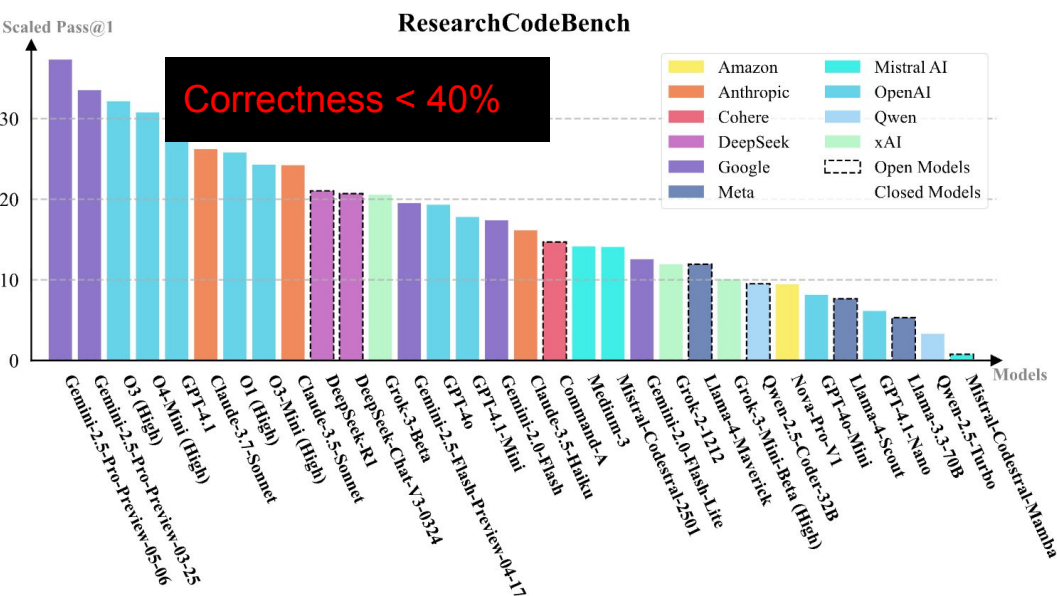
Implementing Ideas



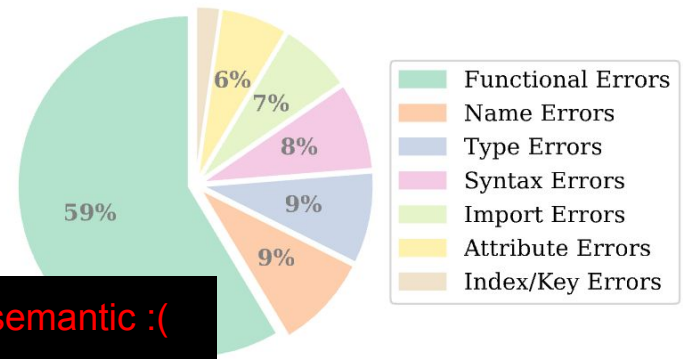
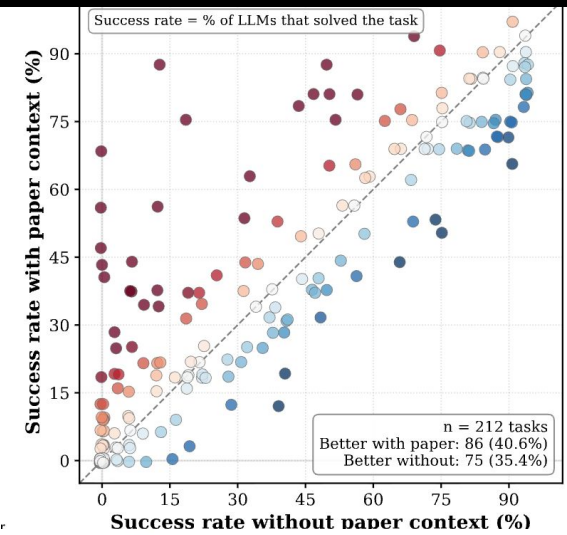
Impact of Paper Context on Per-Task Success Rate



Implementing Ideas



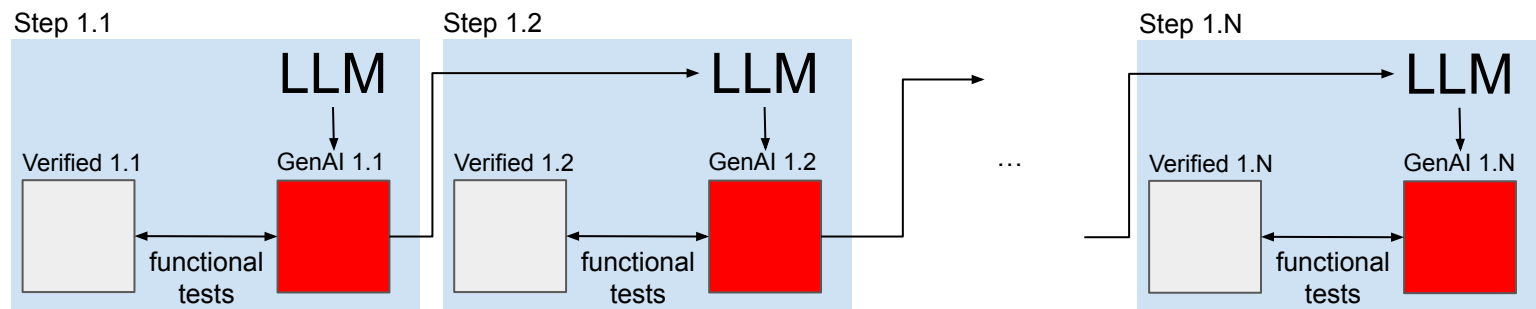
Including the paper can help, sometimes



Errors are mostly semantic :(

Doing Science

Problem 1



Problem 77: Berendsen Thermostat

Main task: Write a script to integrate the Berendsen thermostat and barostat into a molecular dynamics simulation via the velocity Verlet algorithm.

Particles sit in a periodic cubic box, interacting through a truncated-and-shifted Lennard-Jones potential. The thermostat and barostat must each be switchable on/off via their time constants.

Dependencies:

```
python
import math
import numpy as np
import scipy as sp
from scipy.constants import Avogadro
```

Full function signature (`velocityVerlet`, the final assembled function):

Parameters:

`N` : int — number of particles
`xyz` : ndarray — positions, shape (N,3), nm
`v_xyz` : ndarray — velocities, shape (N,3), nm/ps
`L` : float — cubic box side length, nm
`sigma` : float — LJ size parameter, nm
`epsilon` : float — LJ well depth, zJ
`rc` : float — cutoff radius, nm
`m` : float — particle mass, g/mol
`dt` : float — timestep, ps
`tau_T` : float — thermostat time constant, ps (0 = off)
`T_target` : float — target temperature, K
`tau_P` : float — barostat time constant, ps (0 = off)
`P_target` : float — target pressure, bar

Returns:

`xyz_full`, `v_xyz_full`, `L` — updated positions, velocities, box length

Raises:

Exception — if the barostat shrinks the box below $2 \times rc$

77.1 — `wrap`: Wrap to periodic boundaries

Apply periodic boundary conditions to a particle's coordinates in a cubic box.

Background: The unit cell is surrounded by translated copies in every direction to approximate an infinite system — a particle that diffuses across one boundary reappears on the opposite side, so every particle keeps interacting with its neighbors even across box edges.

```
python
def wrap(r, L):
    """Apply periodic boundary conditions to a vector of
    coordinates r for a cubic box of size L.
    Parameters:
    r : The (x, y, z) coordinates of a particle.
    L (float): The length of each side of the cubic box.
    Returns:
    coord: numpy 1d array of floats, the wrapped coordinates
    such that they lie within the cubic box.
    """
    return coord
```

Test cases:

```
python
wrap(np.array([10.5, -1.2, 20.3]), 5.0)
wrap(np.array([10.0, 5.5, -0.1]), 10.0)
wrap(np.array([23.7, -22.1, 14.3]), 10.0)
```

77.2 — `dist`: Minimum image distance

Calculate the minimum image *distance* between two atoms under periodic boundaries.

Background: Implements the minimum image convention — the shortest distance between periodic images of two particles.

```
python
def dist(r1, r2, L):
    """Calculate the minimum image distance between two
    atoms in a periodic cubic system.
    Parameters:
    r1, r2 : (x, y, z) coordinates of the two atoms.
    L (float): The length of the side of the cubic box.
    Returns:
    float: The minimum image distance between the two
    atoms.
    """
    return distance
```

Test cases:

```
python
dist(np.array([2.0,3.0,4.0]), np.array([2.5,3.5,4.5]), 10.0)
dist(np.array([1.0,1.0,1.0]), np.array([9.0,9.0,9.0]), 10.0)
dist(np.array([0.1,0.1,0.1]), np.array([9.9,9.9,9.9]), 10.0)
```


Doing Science

Models	Subproblem	Main Problem
<i>Proprietary Models</i>		
GPT-4o	25.0	1.5
GPT-4-Turbo	22.9	1.5
Claude3.5-Sonnet	26.0	4.6
Claude3-Opus	21.5	1.5
Claude3-Sonnet	17.0	1.5
Gemini 1.5 Pro	21.9	1.5
<i>Open Models</i>		
Llama-3-70B-Chat	14.6	0.0
Mixtral-8x22B-Instruct	16.3	0.0
Deepseek-Coder-v2	21.2	3.1
Qwen2-72B-Instruct	17.0	1.5

w/o scientific background texts

Model	Subproblem		Main Problem	
	Pass@1	Δ	Pass@1	Δ
<i>Proprietary Models</i>				
GPT-4o	35.4	10.4	9.2	7.7
GPT-4-Turbo-2024-04-09	33.7	10.8	9.2	7.7
Claude3.5-Sonnet	35.4	9.4	12.3	7.7
Claude3-Opus	26.7	5.2	4.7	3.0
Claude3-Sonnet	25.7	8.7	4.7	3.0
Gemini 1.5 Pro	30.6	8.7	7.7	6.2
<i>Open Models</i>				
Llama-3-70B-Instruct	20.8	6.3	1.5	1.5
Mixtral-8x22B-Instruct	20.8	4.5	3.1	3.1
Deepseek-Coder-v2	27.1	5.9	4.6	1.5
Qwen2-72B-Instruct	22.2	5.2	4.6	3.1

w/ scientific background texts



Doing Science

Chaining sub problems is hard

Including the paper always helps, still hard

Models	Subproblem	Main Problem
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GPT-4o	25.0	1.5
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w/o scientific background texts

w/ scientific background texts



Version Control

Testing

Environment & Dependencies

Modularity

Documentation

Review